CSA0713-Computer Networks

ASSESSEMENT TOOL -2

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Code of Conduct:

I Sivadharshini.K/192524008(Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

Given:

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Identification of Missing Layers	Correctly identifies both Session and Data Link layers	Identifies one layer correctly	Identifies layers with minor errors	Incorrect identification
Explanation of Functions	Clearly explains all key functions of both layers	Explains most functions correctly	Basic explanation with limited details	Explanation is unclear or incorrect
Analysis of Communication	Clearly analyzes how the layers ensure reliable communication	Good analysis with minor omissions	Limited analysis	No meaningful analysis
Concept Map Accuracy	Concept map is complete and accurate	Minor errors in concept map	Several errors	Incomplete or incorrect concept map

Scoring

16–13: Excellent

12–9: Good

8–5: Satisfactory

4–1: Needs Improvement

This rubric is suitable for evaluating students' understanding of the missing OSI layers and their role in reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supportstransmission.

Given:

Application Layer

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Presentation Layer

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Session Layer

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Transport Layer

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Segment

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Network Layer

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Packet

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Data Link Layer

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Frame

|

Physical Layer

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Bits

Analysis of Data Unit Transformation

OSI Layer	Data Unit	Function
Transport Layer	**Segment**	Divides application data into smaller segments and adds port numbers for end-to-end communication.
Network Layer	**Packet**	Adds IP addresses to segments and determines the best path for data transmission.
Data Link Layer	**Frame**	Encapsulates packets into frames, adds MAC addresses, and performs error detection.
Physical Layer	**Bits**	Converts frames into binary bits (0s and 1s) for transmission through the physical medium.

Analysis

- **Segment → Packet:** The Network layer adds logical (IP) addressing, enabling routing between networks.
- **Packet → Frame:** The Data Link layer adds physical (MAC) addressing and error-checking information for reliable local delivery.
- **Frame → Bits:** The Physical layer converts frames into binary signals (electrical, optical, or wireless) for transmission.
- At the receiver, the process is reversed (**Bits → Frame → Packet → Segment → Data**) through **decapsulation**, ensuring the original data reaches the destination accurately.

Conclusion

The transformation of **Segment → Packet → Frame → Bits** is called **encapsulation**. Each OSI layer adds the necessary information required for reliable, secure, and accurate communication across a network.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery Network → IP Address → Global Delivery
Explain how these relationships ensure complete data delivery across networks.

Given:

Explanation

Data Link Layer → MAC Address → Local Delivery

- The **Data Link Layer** uses **MAC (Media Access Control) addresses**.
- A MAC address uniquely identifies a device within the same local network (LAN).
- It ensures that data frames are delivered to the correct device on the local network.

Network Layer → IP Address → Global Delivery

- The **Network Layer** uses **IP (Internet Protocol) addresses**.

- An IP address identifies devices across different networks.
- Routers use IP addresses to find the best path and deliver packets from the source network to the destination network.

Analysis

- The **IP address** enables data to travel across multiple interconnected networks (global delivery).
- Once the packet reaches the destination network, the **MAC address** ensures it is delivered to the correct device (local delivery).
- Together, the **Network Layer** and **Data Link Layer** provide complete and reliable end-to-end communication.

Conclusion

The **Network Layer** uses **IP addresses** for **global routing**, while the **Data Link Layer** uses **MAC addresses** for **local delivery**. Together, they ensure accurate and reliable data transmission from the source device to the destination device across networks.

5. Develop a concept map for web browsing using OSI layers and analyze how

failure in one layer affects the entire communication process.

Given:

Application Layer

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Web Browser (HTTP/HTTPS)

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Presentation Layer

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Encryption (SSL/TLS), Data Formatting

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Session Layer

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Establishes & Maintains Session

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Transport Layer

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TCP – Reliable Data Transfer

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Network Layer

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IP Addressing & Routing

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Data Link Layer

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MAC Address & Frame Transmission

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Physical Layer

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Bits transmitted through Cable/Wi-Fi/Fiber

Analysis: Effect of Failure in Each Layer

OSI Layer	Failure	Effect on Communication
Application	Browser or HTTP service fails	Web page cannot be requested or displayed.
Presentation	Encryption or data format error	Data cannot be decrypted or understood.
Session	Session establishment fails	Connection is interrupted or terminated.
Transport	TCP failure or packet loss	Data transfer becomes unreliable or incomplete.
Network	IP routing failure	Packets cannot reach the destination.
Data Link	MAC address or frame error	Frames are not delivered correctly on the local network.
Physical	Cable, Wi-Fi, or hardware failure	No data transmission; communication stops completely.

Conclusion

Web browsing depends on all **seven OSI layers** working together. A failure in **any one layer** interrupts the communication process, preventing users from successfully accessing web pages. Each layer performs a specific function that supports reliable, secure, and efficient data transmission across the network.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

Given:

Analysis

- **Physical Layer ✓:** The physical connection (cables, Wi-Fi, switches, etc.) is working correctly.
- **Data Link Layer ✓:** Frames are transmitted successfully, and communication within the local network is functioning.
- **Network Layer ✕:** The problem occurs at the Network layer. Possible causes include:
  - Incorrect IP address
  - Incorrect subnet mask
  - Default gateway misconfiguration
  - Routing failure
- **Transport Layer ✕:** Since the Network layer fails, TCP/UDP cannot establish end-to-end communication.
- **Application Layer ✕:** Applications such as web browsers, email, and file transfer cannot communicate because the lower layers are not delivering data.

How the Concept Map Helps in Troubleshooting

- It identifies the **exact layer** where the problem begins.
- It shows that the **Physical** and **Data Link** layers are functioning correctly.
- It helps network administrators focus on the **Network layer** (IP addressing, routing, gateway, DNS, etc.) instead of checking hardware.
- It reduces troubleshooting time by eliminating unnecessary checks on working layers.

Conclusion

The concept map indicates that the **Network layer is the source of the problem**. Because the OSI model is hierarchical, a failure at the Network layer prevents the **Transport** and **Application** layers from functioning. Thus, the concept map provides a clear and systematic approach to network troubleshooting.